Recursive Computation of Compound Random Variables with a Finite-Mixture Count Distribution

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June 2026

Master's Thesis

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Accepted by the Graduate Department

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The thesis entitled “Recursive Computation of Compound Random Variables with a Finite-Mixture Count Distribution” presented by Robert Righter, a candidate for the degree of Master of Science in Mathematics, has been approved and is worthy of acceptance.

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1. Introduction

Compound random variables have been studied extensively in probability theory and applied fields, as they provide a natural framework for modeling aggregate quantities subject to two layers of randomness: the distribution of the summands and the distribution of the count variable. Classical treatments can be found in [1, 2], where compound distributions are developed in the context of actuarial mathematics and discrete distributions.

While the concept is straightforward, calculating the exact probability distribution of a compound random variable can be a significant computational challenge. Direct methods involving multiple convolutions are often cumbersome and inefficient. A more elegant and powerful approach is the use of recursive methods, which allow for the efficient calculation of the probability $P(S_N = k)$ based on the probabilities of preceding values. This thesis explores one such recursive technique, which is particularly effective when the compounding distribution belongs to a specific class of distributions.

The primary contribution of this work is the extension and simplification of this recursive method to the case where the compounding distribution is a finite mixture of Poisson, Binomial and Negative Binomial distributions. A direct application of the recursive formula to this case leads to a computationally intensive, nested problem, where the mixing weights of the distribution must be recalculated at each step. Additionally, we derive a closed-form expression that computes these recursive weights directly, thereby eliminating some of the nested recursion and simplifying the calculation.

2. Literature Review

Compound random variables can be computed directly via the law of total probability by conditioning on the value of N and convolving the claim size distribution, as described in Ross [1]. However, starting in 1981, alternative approaches were identified that utilize recursive algorithms to compute compound random variables more efficiently.

2.1. The Panjer Recursion

In 1981, Panjer introduced a recursive method for the computation of compound random variables when the counting distribution satisfies a particular recurrence relation [4]. Panjer showed that if the counting distribution N satisfied the recurrence relation,

$$P(N=n)=\left(a+\frac{b}{n}\right)\cdot P(N=n-1), \quad \text{where } n\in\mathbb{Z},\, n>0,\, a,b \text{ are constants,}$$

then $P(S_N=k)$ can be computed with a recursive formula. This eliminated the need to perform explicit convolutions and provided a simpler path for computing $P(S_N=k)$ when the criteria are met.

Panjer's work was extended by Sundt and Jewell, also in 1981 [3]. They proved that only three distribution families satisfy Panjer's recurrence relation, namely, the Poisson, Binomial, and Negative Binomial distribution, also called the $(a,b,0)$ class. They also introduced a relaxed form of the recurrence relation in which the relation holds for $n=2,3,\ldots$ instead of $n=1,2,3,\ldots$, allowing a different initial value at $n=0$. This enabled Panjer's recursive calculations to be extended to counting distributions such as the truncated negative binomial and related zero-modified distributions, collectively called the $(a,b,1)$ class.

Subsequent work by Willmot sought to extend the Panjer framework to accommodate counting distributions formed by mixing members of the $(a,b,0)$ class. In 1993, Willmot derived recursive computation methods for mixed Poisson distributions in which the Poisson parameter is itself a random variable drawn from a continuous mixing distribution [5]. In certain cases, such as the Poisson-Gamma mixture which yields the Negative Binomial, the resulting marginal distribution remains within the Panjer class and therefore satisfies the required recurrence relation. However, this approach is limited to continuous mixtures that collapse into a single known distribution. It does not extend to discrete finite mixtures where the counting distribution is a weighted combination of distributions with fixed parameters, as weighted combination distributions do not satisfy the required recurrence relation.

Panjer recursion remains the standard tool for computation of compound random variables in actuarial science [6]. However, the method is limited to a specific set of counting distributions that satisfy the recurrence relations.

2.2. The Compound Random Variable Identity

In the 2003 eighth edition of his textbook *Introduction to Probability Models*, Ross introduced an identity for compound Poisson random variables and used it to derive a recursive formula for computing their probability mass function [1]. Building on this, Peköz and Ross published a paper in 2004 providing a probabilistic proof of the general identity and applying it to derive recursive formulas for compound random variables when the counting distribution is Poisson, Binomial, Negative Binomial, hypergeometric, logarithmic, or negative hypergeometric [2]. This approach introduces a size-biased auxiliary random variable M, defined as

$$P(M = n) = \frac{n\,P(N = n)}{E[N]}, \quad n = 1, 2, 3, \ldots$$

Unlike the Panjer recursion, this approach does not require the counting distribution to satisfy a specific recurrence relation. However, it does introduce a nuance of recursive depth wherein the distribution of $M - 1$ must be determined at each level of recursion. For the Panjer-class distributions, the distribution of $M - 1$ at each recursion level can be expressed in closed form. In the Poisson case, it is identical to the original distribution; in the Binomial and Negative Binomial cases, it is the same family with a shifted parameter. Peköz and Ross applied the identity to each of these distributions individually but did not explore the case where the counting distribution is a finite mixture.

3. Background

Definition 1. The expected value of a discrete random variable, X, is denoted by $E(X)$, and given by

$$E[X] = \sum_{\{x\,:\,f(x)>0\}} x \cdot f(x),$$

where $f(x)$ is the probability mass function of X.

Definition 2. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables. Let N be a nonnegative integer valued random variable that is independent of the sequence $X_1, X_2, \ldots, X_n$. Then,

$$S_N = \sum_{i=1}^{N} X_i$$

is a *compound random variable* [1].

Definition 3. In a compound random variable, the distribution of N is called the *compounding distribution.*

Theorem 1 (Compound Random Variable Identity [1, 2])**.** *Let M be a random variable that is independent of the sequence of random variables $X_1, X_2, \ldots, X_n$ that forms a compound random variable and M satisfies*

$$P(M = n) = \frac{n\,P(N = n)}{E[N]}, \qquad n = 1, 2, 3, \ldots$$

Then, for any function h,

$$E(S_N \cdot h(S_N)) = E[N] \cdot E(X_1 \cdot h(S_M))\,.$$

Proof.

$$\begin{aligned} E(S_N \cdot h(S_N)) &= E\left[\sum_{i=1}^{N} X_i\, h(S_N)\right] \\ &= E\left[E\left(\sum_{i=1}^{N} X_i\, h(S_N) \;\middle|\; N = n\right)\right] \end{aligned}$$

$$= \sum_{n=0}^{\infty} E\left(\sum_{i=1}^{n} X_i\, h(S_n) \,\middle|\, N = n\right) P(N = n)$$

$$= \sum_{n=0}^{\infty} E\left(\sum_{i=1}^{n} X_i\, h(S_n)\right) P(N = n)$$

$$= \sum_{n=0}^{\infty} \sum_{i=1}^{n} E(X_i\, h(S_n))\, P(N = n).$$

Since $X_1, X_2, \ldots$ are independent and identically distributed, we have

$$E(X_i \cdot h(S_N)) = E(X_j \cdot h(S_N))\,, \qquad \forall\, i, j \in \{1, 2, \ldots, n\}.$$

So,

$$\begin{aligned} E(S_N \cdot h(S_N)) &= \sum_{n=0}^{\infty} \sum_{i=1}^{n} E(X_1\, h(S_n))\, P(N = n) \\ &= \sum_{n=1}^{\infty} n \cdot E(X_1\, h(S_n))\, P(N = n). \end{aligned}$$

Then, by substitution of $P(M = n) = \dfrac{n\, P(N = n)}{E[N]}$, we have

$$= \sum_{n=0}^{\infty} E[N] \cdot E(X_1\, h(S_n))\, P(M = n)$$

$$= E[N] \sum_{n=0}^{\infty} E(X_1\, h(S_n))\, P(M = n)$$

$$= E[N] \sum_{n=0}^{\infty} E(X_1\, h(S_n) \mid M = n)\, P(M = n)$$

$$= E[N] \sum_{n=0}^{\infty} E(X_1\, h(S_M) \mid M = n)\, P(M = n)$$

$$= E[N]\, E(E(X_1\, h(S_M) \mid M = n))$$

$$= E[N]\, E(X_1\, h(S_M))\,. \qquad \square$$

Corollary 1 ([2]). *Let S_N be a compound random variable and suppose $X_1, X_2, \ldots, X_n$ are positive integer valued random variables. Suppose $P(X_1 = i) = \alpha_i$, $i > 0$, and M is a random variable that is independent of the sequence $X_1, X_2, \ldots, X_n$ and satisfies*

$$P(M = n) = \frac{n\,P(N = n)}{E[N]}.$$

Then,

$$P(S_N = 0) = P(N = 0) \quad \textit{and} \quad P(S_N = k) = \frac{1}{k}\,E[N] \sum_{i=1}^{k} i\,\alpha_i\,P(S_{M-1} = k - i).$$

Proof. For a fixed k, let

$$h(x) = \begin{cases} 1 & \text{if } x = k, \\ 0 & \text{otherwise,} \end{cases}$$

and note that

$$S_N \cdot h(S_N) = \begin{cases} k & \text{if } S_N = k, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$E(S_N \cdot h(S_N)) = k \cdot P(S_N = k).$$

Now, by the Compound Random Variable Identity, $E(S_N \cdot h(S_N)) = E[N] \cdot E(X_1 \cdot h(S_M))$. So,

$$\begin{aligned} k \cdot P(S_N = k) &= E[N] \cdot E(X_1 \cdot h(S_M)) \\ &= E[N] \cdot E(E(X_1 \cdot h(S_M) \mid X_1 = j)) \\ &= E[N] \sum_{j=1}^{\infty} E(X_1 \cdot h(S_M) \mid X_1 = j) \cdot P(X_1 = j) \\ &= E[N] \sum_{j=1}^{\infty} E(X_1 \cdot h(S_M) \mid X_1 = j) \cdot \alpha_j \\ &= E[N] \sum_{j=1}^{\infty} E(j \cdot h(S_M) \mid X_1 = j) \cdot \alpha_j \end{aligned}$$

$$= E[N]\sum_{j=1}^{\infty} j \cdot E(h(S_M) \mid X_1 = j) \cdot \alpha_j.$$

Now,

$$E(h(S_M) \mid X_1 = j) = \sum_{S'_M} h(S'_M)\, f_{S_M|X_1}(S'_M \mid j) = P(S_M = k \mid X_1 = j).$$

And,

$$\begin{aligned} P(S_M = k \mid X_1 = j) &= P\left(\sum_{i=1}^{m} X_i = k \,\middle|\, X_1 = j\right) \\ &= P\left(j + \sum_{i=2}^{m} X_i = k\right) = P\left(\sum_{i=2}^{m} X_i = k - j\right). \end{aligned}$$

Letting $\ell = i - 1$,

$$= P\left(\sum_{\ell=1}^{m-1} X_\ell = k - j\right) = P(S_{M-1} = k - j).$$

So,

$$k \cdot P(S_N = k) = E[N]\sum_{j=1}^{\infty} j \cdot \alpha_j \cdot P(S_{M-1} = k - j).$$

Note, when $j > k$, $P(S_{M-1} = k - j) = 0$. So,

$$k \cdot P(S_N = k) = E[N]\sum_{j=1}^{k} j \cdot \alpha_j \cdot P(S_{M-1} = k - j),$$

and therefore,

$$P(S_N = k) = \frac{1}{k} \cdot E[N]\sum_{j=1}^{k} j \cdot \alpha_j \cdot P(S_{M-1} = k - j).$$

Notice, by definition, for $N \geq 1$,

$$P(S_N = 0) = P\left(\sum_{j=i}^{N} X_i = 0\right) = 0,$$

so $P(S_N = 0) = P(N = 0)$. $\square$

4. Examples

For each of the distributions below, let $S_N = \sum_{i=1}^{N} X_i$ be a compound random variable where

$$P(X_1 = 1) = 0.05, \quad P(X_1 = 2) = 0.4, \quad P(X_1 = 3) = 0.1,$$

$$P(X_1 = 4) = 0.25, \quad P(X_1 = 5) = 0.2.$$

Find $P(S_N = 4)$.

Example 1 (N in the Poisson Distribution)**.** Let N be a random variable of the Poisson distribution with parameter $\lambda = 3$. So the probability mass function for the distribution of $N^{(s)}$ at recursion level s is

$$P^{(s)}(N = n) = \frac{e^{-\lambda}\,\lambda^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

We claim that $E[N^{(s)}] = \lambda$, where $E[N^{(s)}]$ is the expected value of the distribution of N at the level of recursion s.

Proof.

$$\begin{aligned} E[N^{(s)}] &= \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\,\lambda^k}{k!} = 0 + \sum_{k=1}^{\infty} k \cdot \frac{e^{-\lambda}\,\lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\,\lambda^k}{(k-1)!} \\ &= e^{-\lambda} \cdot \lambda \cdot \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}. \end{aligned}$$

Now let $j = k - 1$. So when $k = 1$, $j = 0$, and as $k \to \infty$, $j \to \infty$. So,

$$E[N^{(s)}] = \lambda \cdot e^{-\lambda} \cdot \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}.$$

We recognize $\sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$ as the Taylor series for e^{λ}. So,

$$E[N^{(s)}] = \lambda \cdot e^{-\lambda} \cdot e^{\lambda} = \lambda. \qquad \square$$

Therefore, $E[N^{(s)}] = \lambda = 3$.

Before applying Corollary 1 to compute $P(S_N = 4)$, consider an alternate, more intuitive approach. This method utilizes the law of total probability and directly sums the probability of every outcome that yields the target sum.

Recall the *law of total probability*: to calculate $E[X]$ we may take a weighted average of the conditional expected value of X given $Y = y$, each term $E[X \mid Y = y]$ being weighted by $P(Y = y)$ [1, p. 107]. Written formally,

$$E[X] = \sum_y E[X \mid Y = y] \cdot P(Y = y).$$

Proof.

$$\begin{aligned}
\sum_y E[X \mid Y = y] \cdot P(Y = y) &= \sum_y \sum_x x\, P(X = x \mid Y = y) \cdot P(Y = y) \\
&= \sum_y \sum_x x \cdot \frac{P(X = x,\, Y = y)}{P(Y = y)} \cdot P(Y = y) \\
&= \sum_y \sum_x x \cdot P(X = x,\, Y = y) \\
&= \sum_x x \sum_y P(X = x,\, Y = y) \\
&= \sum_x x \cdot P(X = x) \;=\; E[X].
\end{aligned}$$

□

Applying this formula to our specific example, we must determine all of the possible ways for the sum to equal 4. The full list of possibilities is:

When $N = 1$, we must have $X_1 = 4$.

When $N = 2$, we must have $(X_1 = 2,\, X_2 = 2)$, or $(X_1 = 1,\, X_2 = 3)$, or $(X_1 = 3,\, X_2 = 1)$.

When $N = 3$, we must have $(X_1 = 2,\, X_2 = 1,\, X_3 = 1)$, or $(X_1 = 1,\, X_2 = 2,\, X_3 = 1)$, or $(X_1 = 1,\, X_2 = 1,\, X_3 = 2)$.

When $N = 4$, we must have $(X_1 = 1,\, X_2 = 1,\, X_3 = 1,\, X_4 = 1)$.

We know that

$$
\begin{aligned}
P(N=1) &= \frac{e^{-3}\,3^1}{1!} = e^{-3}\cdot 3,\\
P(N=2) &= \frac{e^{-3}\,3^2}{2!} = e^{-3}\cdot \frac{9}{2},\\
P(N=3) &= \frac{e^{-3}\,3^3}{3!} = e^{-3}\cdot \frac{27}{6},\\
P(N=4) &= \frac{e^{-3}\,3^4}{4!} = e^{-3}\cdot \frac{81}{24}.
\end{aligned}
$$

So, we apply $P(S_N=4)=\sum_{n=1}^{4} P(S_N=4 \mid N=n)\,P(N=n)$ and we have,

$$
\begin{aligned}
P(S_N=4) = \big(e^{-3}\cdot 3\big)(0.25) &+ \Big(e^{-3}\cdot \tfrac{9}{2}\Big)\big((0.4)^2+2(0.05\cdot 0.1)\big)\\
&+ \Big(e^{-3}\cdot \tfrac{27}{6}\Big)3\,(0.05)^2(0.4) + \Big(e^{-3}\cdot \tfrac{81}{24}\Big)(0.05)^4\\
&= e^{-3}\left[(3\cdot 0.25)+\frac{9}{2}\big((0.4)^2+2(0.05\cdot 0.1)\big)+\frac{27}{6}\cdot 3\,(0.05)^2(0.4)+\frac{81}{24}\,(0.05)^4\right]\\
&= e^{-3}\cdot 1.528521094.
\end{aligned}
$$

With this intuitive example in mind, we now calculate this same compound random variable using the formula from Corollary 1.

So we have,

$$
\begin{aligned}
P(M^{(s)}-1=n) = P(M^{(s)}=n+1) &= \frac{(n+1)\,P^{(s)}(N=n+1)}{E[N^{(s)}]}\\
&= \frac{n+1}{\lambda}\cdot\frac{e^{-\lambda}\,\lambda^{n+1}}{(n+1)!} = \frac{e^{-\lambda}\,\lambda^{n}}{n!}.
\end{aligned}
$$

Therefore, we see that $P^{(s+1)}(N=n)$ is also the Poisson distribution with parameter λ. Since the distribution is unchanged at each recursion level, we have

$$
P^{(s)}(S_N=k) = \frac{\lambda}{k}\sum_{i=1}^{k} i\,\alpha_i\cdot P^{(s+1)}(S_N=k-i),
$$

which, because $P^{(s)}=P^{(s+1)}$ for all $s\geq 0$, simplifies to

$$
P(S_N=k) = \frac{\lambda}{k}\sum_{i=1}^{k} i\,\alpha_i\cdot P(S_N=k-i).
$$

Now we calculate $P(S_N = 4)$.

$$P(S_N = 0) = P(N = 0) = \frac{e^{-\lambda}\,\lambda^0}{0!} = e^{-\lambda} = e^{-3}.$$

$$P(S_N = 1) = \frac{3}{1}\big[1 \cdot (0.05) \cdot P(S_N = 0)\big] = 3 \cdot (0.05) \cdot e^{-3} = 0.15\, e^{-3}.$$

$$\begin{aligned} P(S_N = 2) &= \frac{3}{2}\big[(0.05) \cdot P(S_N = 1) + 2 \cdot (0.4) \cdot P(S_N = 0)\big] \\ &= \frac{3}{2}\big[(0.05)(0.15\, e^{-3}) + (0.8)\, e^{-3}\big] = 1.21125\, e^{-3}. \end{aligned}$$

$$\begin{aligned} P(S_N = 3) &= \frac{3}{3}\big[(0.05) \cdot P(S_N = 2) + (0.8) \cdot P(S_N = 1) + 3 \cdot (0.1) \cdot P(S_N = 0)\big] \\ &= (0.05)(1.21125\, e^{-3}) + (0.8)(0.15\, e^{-3}) + (0.3)\, e^{-3} = 0.4805625\, e^{-3}. \end{aligned}$$

$$\begin{aligned} P(S_N = 4) &= \frac{3}{4}\big[0.05 \cdot P(S_N = 3) + 0.8 \cdot P(S_N = 2) + 0.3 \cdot P(S_N = 1) + 4 \cdot (0.25) \cdot P(S_N = 0)\big] \\ &= \frac{3}{4}\big[0.05(0.4805625\, e^{-3}) + 0.8(1.21125\, e^{-3}) + 0.3(0.15\, e^{-3}) + e^{-3}\big] \\ &= 1.528521094\, e^{-3}. \end{aligned}$$

Note that this result agrees with the value $e^{-3} \cdot 1.528521094$ obtained above via the law of total probability, confirming the two methods are consistent.

Example 2 (N in the Negative Binomial Distribution)**.** Let N be a random variable of the negative binomial distribution with parameters $r = 6$ and $p = 0.6$. The probability mass function for the distribution of $N^{(s)}$ at recursion level s is

$$P^{(s)}(N = n) = \binom{n + r + s - 1}{n} p^{r+s}(1 - p)^n, \qquad n \geq 0.$$

We claim that $E[N^{(s)}] = \dfrac{(r + s)(1 - p)}{p}$.

Proof.

$$E[N^{(s)}] = \sum_{k=0}^{\infty} k \binom{k + r + s - 1}{k} p^{r+s}(1 - p)^k = \sum_{k=1}^{\infty} k \binom{k + r + s - 1}{k} p^{r+s}(1 - p)^k.$$

Recalling that $\binom{a}{b} = \dfrac{a!}{b!\,(a-b)!}$, we see that

$$k\binom{k+r+s-1}{k} = k \cdot \frac{(k+r+s-1)!}{k!\,(r+s-1)!} = \frac{(k+r+s-1)!}{(k-1)!\,(r+s-1)!} = (r+s)\binom{k+r+s-1}{k-1}.$$

So we have,

$$E[N^{(s)}] = (r+s)(1-p)\,p^{r+s}\sum_{k=1}^{\infty}\binom{k+r+s-1}{k-1}(1-p)^{k-1}.$$

Let $m = k-1$. By substitution,

$$E[N^{(s)}] = (r+s)(1-p)\,p^{r+s}\sum_{m=0}^{\infty}\binom{m+(r+s)}{m}(1-p)^{m}.$$

Recalling the negative binomial theorem, $(1+x)^{-t} = \sum_{i=0}^{\infty}\binom{-t}{i}x^i$, and applying it, we see that

$$\sum_{m=0}^{\infty}\binom{m+r+s}{m}(1-p)^m = p^{-(r+s+1)}.$$

So by substitution,

$$E[N^{(s)}] = (r+s)(1-p)\,p^{r+s}\cdot p^{-(r+s+1)} = \frac{(r+s)(1-p)}{p}. \qquad \square$$

So we have,

$$\begin{aligned}
P^{(s)}(M^{(s)}-1=n) &= P^{(s)}(M^{(s)}=n+1)\\
&= \frac{(n+1)\,P^{(s)}(N=n+1)}{E[N^{(s)}]}\\
&= \binom{n+r+s}{n+1}p^{r+s}(1-p)^{n+1}\cdot\frac{(n+1)}{(r+s)(1-p)/p}\\
&= \frac{(n+1)\,p}{(r+s)(1-p)}\cdot\frac{(n+r+s)!}{(n+1)!\,(r+s-1)!}\cdot p^{r+s}(1-p)^{n+1}\\
&= \frac{(n+r+s)!}{n!\,(r+s)!}\cdot p^{r+s+1}(1-p)^n\\
&= \binom{n+(r+s+1)-1}{n}p^{r+s+1}(1-p)^n = P^{(s+1)}(N=n).
\end{aligned}$$

Therefore, we see that $P^{(s)}(M^{(s)}-1=n)$ is also the negative binomial distribution but at the next recursion level $(s+1)$. So, we have

$$P^{(s)}(S_N=k)=\frac{(r+s)(1-p)}{k\,p}\sum_{i=1}^{k} i\,\alpha_i\cdot P^{(s+1)}(S_N=k-i).$$

Now we can begin the computation. With $r=6$, $p=0.6$ (so $1-p=0.4$), at level $s=0$ the coefficient is $\dfrac{r(1-p)}{p}=4$, giving

$$P^{(0)}(S_N=k)=\frac{4}{k}\sum_{i=1}^{k} i\,\alpha_i\cdot P^{(1)}(S_N=k-i).$$

Base cases $P^{(s)}(S_N=0)=p^{r+s}$:

$$P^{(0)}(S_N=0)=(0.6)^6=0.046656.$$

$$P^{(1)}(S_N=0)=(0.6)^7=0.0279936.$$

$$P^{(2)}(S_N=0)=(0.6)^8=0.01679616.$$

$$P^{(3)}(S_N=0)=(0.6)^9=0.010077696.$$

$$P^{(4)}(S_N=0)=(0.6)^{10}=0.0060466176.$$

Computing $P^{(s)}(S_N=1)$:

$$P^{(0)}(S_N=1)=\frac{(r)(1-p)}{1\cdot p}\big[0.05\cdot P^{(1)}(S_N=0)\big]=(0.2)(0.0279936)=0.00559872.$$

$$P^{(1)}(S_N=1)=\frac{(r+1)(1-p)}{1\cdot p}\big[0.05\cdot P^{(2)}(S_N=0)\big]=\tfrac{7\cdot 0.4}{0.6}(0.05)(0.01679616)=0.003919104.$$

$$P^{(2)}(S_N=1)=\frac{(r+2)(1-p)}{1\cdot p}\big[0.05\cdot P^{(3)}(S_N=0)\big]=\tfrac{8\cdot 0.4}{0.6}(0.05)(0.010077696)=0.0026873856.$$

$$P^{(3)}(S_N=1)=\frac{(r+3)(1-p)}{1\cdot p}\big[0.05\cdot P^{(4)}(S_N=0)\big]=\tfrac{9\cdot 0.4}{0.6}(0.05)(0.0060466176)=0.00181398528.$$

Computing $P^{(s)}(S_N=2)$:

$$\begin{aligned}P^{(1)}(S_N=2)&=\frac{(r+1)(1-p)}{2\,p}\big[(0.05)\cdot P^{(2)}(S_N=1)+(0.8)\cdot P^{(2)}(S_N=0)\big]\\&=\frac{7\cdot 0.4}{2\cdot 0.6}\big[(0.05)(0.0026873856)+(0.8)(0.01679616)\big]=0.03166636032.\end{aligned}$$

$$
\begin{aligned}
P^{(2)}(S_N = 2) &= \frac{(r+2)(1-p)}{2\,p}\big[(0.05)\cdot P^{(3)}(S_N = 1) + (0.8)\cdot P^{(3)}(S_N = 0)\big] \\
&= \frac{8\cdot 0.4}{2\cdot 0.6}\big[(0.05)(0.00181398528) + (0.8)(0.010077696)\big] = 0.021740949504.
\end{aligned}
$$

Computing $P^{(1)}(S_N = 3)$:

$$
\begin{aligned}
P^{(1)}(S_N = 3) &= \frac{(r+1)(1-p)}{3\,p}\big[(0.05)\cdot P^{(2)}(S_N = 2) + (0.8)\cdot P^{(2)}(S_N = 1) + (0.3)\cdot P^{(2)}(S_N = 0)\big] \\
&= \frac{7\cdot 0.4}{3\cdot 0.6}\big[(0.05)(0.021740949504) + (0.8)(0.0026873856) + (0.3)(0.01679616)\big] \\
&= 0.01287347282.
\end{aligned}
$$

Computing $P^{(0)}(S_N = 4)$:

$$
\begin{aligned}
P^{(0)}(S_N = 4) &= \frac{r(1-p)}{4\,p}\big[0.05\cdot P^{(1)}(S_N = 3) + 0.8\cdot P^{(1)}(S_N = 2) + 0.3\cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\
&= \frac{6\cdot 0.4}{4\cdot 0.6}\big[(0.05)(0.01287347282) + (0.8)(0.03166636032) \\
&\qquad\qquad + (0.3)(0.003919104) + 0.0279936\big] \\
&= \boxed{0.05514609310}.
\end{aligned}
$$

Example 3 (N in the Binomial Distribution)**.** Let N be a random variable of the Binomial distribution with parameters $r = 6$ and $p = 0.6$. The probability mass function for the distribution of $N^{(s)}$ at recursion level s is

$$
P^{(s)}(N = n) = \binom{r-s}{n} p^n (1-p)^{(r-s)-n}, \qquad n \geq 0.
$$

We claim that $E[N^{(s)}] = (r-s)\,p$.

Proof.

$$
E[N^{(s)}] = \sum_{k=0}^{\infty} k\binom{r-s}{k} p^k (1-p)^{(r-s)-k} = \sum_{k=1}^{\infty} k\binom{r-s}{k} p^k (1-p)^{(r-s)-k}.
$$

Notice that

$$k\binom{r-s}{k} = k \cdot \frac{(r-s)!}{k!\,(r-s-k)!} = \frac{(r-s)!}{(k-1)!\,(r-s-k)!} = (r-s)\binom{r-s-1}{k-1}.$$

By substitution,

$$E[N^{(s)}] = (r-s)\,p\sum_{k=1}^{\infty}\binom{r-s-1}{k-1}p^{k-1}(1-p)^{(r-s)-k}.$$

We let $m = k-1$. So we have,

$$E[N^{(s)}] = (r-s)\,p\sum_{m=0}^{\infty}\binom{r-s-1}{m}p^{m}(1-p)^{(r-s-1)-m}.$$

And by the binomial theorem,

$$\sum_{m=0}^{\infty}\binom{r-s-1}{m}p^{m}(1-p)^{(r-s-1)-m} = \big[p+(1-p)\big]^{r-s-1} = 1.$$

By substitution we have $E[N^{(s)}] = (r-s)\,p$. $\square$

So, for this example at $s=0$, $E[N^{(0)}] = r\,p = 3.6$. We have,

$$\begin{aligned} P^{(s)}(M^{(s)}-1=n) = P^{(s)}(M^{(s)}=n+1) &= \frac{(n+1)\,P^{(s)}(N=n+1)}{E[N^{(s)}]} \\ &= \binom{r-s}{n+1}p^{n+1}(1-p)^{(r-s)-n-1}\cdot\frac{n+1}{(r-s)\,p} \\ &= \frac{(r-s)!}{(n+1)!\,(r-s-n-1)!}\,p(1-p)^{(r-s-1)-n}\cdot\frac{n+1}{(r-s)\,p} \\ &= \frac{(r-s-1)!}{n!\,(r-s-1-n)!}\cdot p^{n}(1-p)^{(r-s-1)-n} \\ &= \binom{r-s-1}{n}p^{n}(1-p)^{(r-s-1)-n} = P^{(s+1)}(N=n). \end{aligned}$$

Therefore, we see that $P^{(s)}(M^{(s)}-1=n)$ is also the binomial distribution but at the next recursion level $(s+1)$, which corresponds to a parameter change to $(r-s-1)$. So, we have

$$P^{(s)}(S_N=k) = \frac{(r-s)\,p}{k}\sum_{i=1}^{k} i\,\alpha_i\cdot P^{(s+1)}(S_N=k-i).$$

Now we can begin the computation to find $P(S_N=4)$.

Base cases $P^{(s)}(S_N = 0) = (1-p)^{r-s}$:

$$P^{(1)}(S_N = 0) = P^{(1)}(N = 0) = (0.4)^5 = 0.01024.$$
$$P^{(2)}(S_N = 0) = P^{(2)}(N = 0) = (0.4)^4 = 0.0256.$$
$$P^{(3)}(S_N = 0) = P^{(3)}(N = 0) = (0.4)^3 = 0.064.$$

From Corollary 1, $P^{(0)}(S_N = 0) = P^{(0)}(N = 0) = (0.4)^6 = 0.004096$.

$$P^{(0)}(S_N = 1) = \frac{r\,p}{1}\big[1 \cdot (0.05) \cdot P^{(1)}(S_N = 0)\big] = 6(0.6)(0.05)(0.01024) = 0.0018432.$$
$$P^{(3)}(S_N = 1) = \frac{(r-3)\,p}{1}\big[0.05 \cdot P^{(4)}(S_N = 0)\big] = 3(0.6)(0.05)(0.4)^2 = 0.0144.$$
$$P^{(2)}(S_N = 1) = \frac{(r-2)\,p}{1}\big[0.05 \cdot P^{(3)}(S_N = 0)\big] = 4(0.6)(0.05)(0.064) = 0.00768.$$
$$P^{(1)}(S_N = 1) = \frac{(r-1)\,p}{1}\big[0.05 \cdot P^{(2)}(S_N = 0)\big] = 5(0.6)(0.05)(0.0256) = 0.00384.$$

$$\begin{aligned} P^{(2)}(S_N = 2) &= \frac{(r-2)\,p}{2}\big[0.05 \cdot P^{(3)}(S_N = 1) + 0.8 \cdot P^{(3)}(S_N = 0)\big] \\ &= \frac{4(0.6)}{2}\big[0.05(0.0144) + 0.8(0.064)\big] = 0.062304. \\ P^{(1)}(S_N = 2) &= \frac{(r-1)\,p}{2}\big[0.05 \cdot P^{(2)}(S_N = 1) + 0.8 \cdot P^{(2)}(S_N = 0)\big] \\ &= \frac{5(0.6)}{2}\big[0.05(0.00768) + 0.8(0.0256)\big] = 0.031296. \end{aligned}$$

$$\begin{aligned} P^{(1)}(S_N = 3) &= \frac{(r-1)\,p}{3}\big[(0.05) \cdot P^{(2)}(S_N = 2) + (0.8) \cdot P^{(2)}(S_N = 1) + (0.3) \cdot P^{(2)}(S_N = 0)\big] \\ &= \frac{5(0.6)}{3}\big[(0.05)(0.062304) + (0.8)(0.00768) + (0.3)(0.0256)\big] = 0.016937200. \end{aligned}$$

$$\begin{aligned} P^{(0)}(S_N = 4) &= \frac{r\,p}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\ &= \frac{6(0.6)}{4}\big[(0.05)(0.016937200) + (0.8)(0.031296) + (0.3)(0.00384) + 0.01024\big] \\ &= \boxed{0.033548184}. \end{aligned}$$

5. Computation with a Finite Mixture of Poisson Distributions

We will now consider the computation of compound random variable probabilities when the counting distribution is a mixture of Poisson distributions.

Example 4 (N in a Finite Mixture of Two Poisson Distributions)**.** Consider a discrete mixture distribution where the event is generated by one of two random processes. The event is either generated by process one, which follows a Poisson distribution with mean λ_1, or process two, which follows a Poisson distribution with mean λ_2. The probability of process one is β_1, and the probability of process two is β_2, where $\beta_1 + \beta_2 = 1$. Let the parameters of this distribution be $\lambda_1, \lambda_2, \beta_1, \beta_2$.

So, we have,

$$P^{(s)}(N = n) = \beta_1^{(s)} \cdot \frac{e^{-\lambda_1}\,\lambda_1^n}{n!} + \beta_2^{(s)} \cdot \frac{e^{-\lambda_2}\,\lambda_2^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

We claim that $E[N^{(s)}] = \beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2$.

Proof.

$$\begin{aligned} E[N^{(s)}] &= \sum_{k=0}^{\infty} k \left[\beta_1^{(s)} \cdot \frac{e^{-\lambda_1}\,\lambda_1^k}{k!} + \beta_2^{(s)} \cdot \frac{e^{-\lambda_2}\,\lambda_2^k}{k!}\right] \\ &= \beta_1^{(s)} \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda_1}\,\lambda_1^k}{k!} + \beta_2^{(s)} \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda_2}\,\lambda_2^k}{k!}. \end{aligned}$$

Recall from Example 1 that we showed $\sum_{k=0}^{\infty} k \cdot \dfrac{e^{-\lambda_i}\,\lambda_i^k}{k!} = \lambda_i$. So, by substitution we have $E[N^{(s)}] = \beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2$. □

Proposition 1. *The probability mass function* $P(N = n) = \beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^n}{n!} + \beta_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^n}{n!}$ *is not a member of the Panjer class.*

Proof. Suppose for a contradiction that $P(N=n) = \beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^n}{n!} + \beta_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^n}{n!}$ is a member of the Panjer class. Therefore, the function must satisfy the Panjer recurrence relation,

$$P(N=n) = \left(a + \frac{b}{n}\right) \cdot P(N=n-1), \qquad n \in \mathbb{Z},\ n > 0,\ a, b \text{ are constants.}$$

So,

$$\frac{P(N=n)}{P(N=n-1)} = \left(a + \frac{b}{n}\right), \qquad \forall\, n \in \mathbb{Z} \text{ where } n > 0,\ a, b \text{ are constants.}$$

Then,

$$\frac{\beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^n}{n!} + \beta_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^n}{n!}}{\beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^{n-1}}{(n-1)!} + \beta_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^{n-1}}{(n-1)!}} = \left(a + \frac{b}{n}\right), \qquad \forall\, n \in \mathbb{Z} \text{ where } n > 0,\ a, b \text{ are constants.}$$

We evaluate the ratio for $n = 1$, $n = 2$, $n = 3$, and $n = 4$. Let $\beta_1 = 0.5$, $\beta_2 = 0.5$, $\lambda_1 = 1$, $\lambda_2 = 2$.

When $n = 1$ we have,

$$= \frac{e^{-1} + e^{-2} \cdot 2}{e^{-1} + e^{-2}} = (a + b), \qquad \text{where } a, b \text{ are constants,}$$

$$= \frac{e+2}{e+1} = a + b, \qquad \text{where } a, b \text{ are constants.}$$

When $n = 2$ we have,

$$= \frac{\dfrac{e^{-1}}{2} + \dfrac{e^{-2} \cdot 4}{2}}{e^{-1} + e^{-2} \cdot 2} = \frac{2a+b}{2}, \qquad \text{where } a, b \text{ are constants,}$$

$$= \frac{1}{2} \cdot \frac{e+4}{e+2} = \frac{2a+b}{2}, \qquad \text{where } a, b \text{ are constants,}$$

$$\frac{e+4}{e+2} = 2a + b, \qquad \text{where } a, b \text{ are constants.}$$

When $n = 3$ we have,

$$= \frac{\dfrac{e^{-1}}{6} + \dfrac{e^{-2} \cdot 8}{6}}{\dfrac{e^{-1}}{2} + \dfrac{e^{-2} \cdot 4}{2}} = \frac{3a+b}{3}, \qquad \text{where } a, b \text{ are constants,}$$

$$\frac{1}{3} \cdot \frac{e+8}{e+4} = \frac{3a+b}{3}, \qquad \text{where } a, b \text{ are constants,}$$

$$\frac{e+8}{e+4} = 3a + b, \qquad \text{where } a, b \text{ are constants.}$$

Solving for a we have,

$$\begin{aligned}\frac{e+2}{e+1} &= a+b,\\ a &= \frac{e+2}{e+1} - b.\end{aligned}$$

Solving for b we have,

$$\begin{aligned}\frac{e+4}{e+2} &= 2a+b,\\ b &= \frac{e+4}{e+2} - 2a.\end{aligned}$$

And by substitution,

$$\begin{aligned}a &= \frac{e+2}{e+1} - \left(\frac{e+4}{e+2} - 2a\right)\\ a &= \frac{e+2}{e+1} - \left(\frac{e+4}{e+2}\right) + 2a\\ a &= \frac{e+4}{e+2} - \frac{e+2}{e+1} = \frac{(e^2+5e+4)-(e^2+4e+4)}{e^2+3e+2}\\ a &= \frac{e}{e^2+3e+2}.\end{aligned}$$

And also by substitution,

$$\begin{aligned}b &= \frac{e+4}{e+2} - 2\left(\frac{e}{e^2+3e+2}\right)\\ b &= \frac{e+4}{e+2} - \frac{2e}{(e+1)(e+2)}\\ b &= \frac{(e+1)(e+4)-2e}{(e+1)(e+2)}\\ b &= \frac{e^2+3e+4}{(e+1)(e+2)}.\end{aligned}$$

So we must have,

$$\frac{e+8}{e+4} = 3a+b.$$

Then by substitution,

$$\begin{aligned}\frac{e+8}{e+4} &= \left(3\cdot\frac{e}{e^2+3e+2} + \frac{e^2+3e+4}{(e+1)(e+2)}\right)\\ \frac{e+8}{e+4} &= \frac{e^2+6e+4}{e^2+3e+2}\end{aligned}$$

$$e^3 + 11e^2 + 26e + 16 = e^3 + 10e^2 + 28e + 16$$
$$e^2 = 2e.$$

This implies $e = 2$, which contradicts the known irrational value of e. Therefore, the Panjer recurrence relation does not hold, and $P(N = n) = \beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^n}{n!} + \beta_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^n}{n!}$ is not a member of the Panjer class. $\square$

Having shown that $P(N = n)$ is not in the Panjer class and is therefore not eligible for Panjer recursion, we will seek to apply the Peköz and Ross method.

So, we have,

$$\begin{aligned}
P(M^{(s)} - 1 = n) = P(M^{(s)} = n + 1) &= \frac{(n+1)\,P^{(s)}(N = n+1)}{E[N^{(s)}]} \\
&= \frac{(n+1)\left[\beta_1^{(s)} \cdot \dfrac{e^{-\lambda_1}\lambda_1^{n+1}}{(n+1)!} + \beta_2^{(s)} \cdot \dfrac{e^{-\lambda_2}\lambda_2^{n+1}}{(n+1)!}\right]}{\beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2} \\
&= \frac{\beta_1^{(s)}\lambda_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^{n}}{n!} + \beta_2^{(s)}\lambda_2 \cdot \dfrac{e^{-\lambda_2}\lambda_2^{n}}{n!}}{\beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2} \\
&= \frac{\beta_1^{(s)}\lambda_1}{\beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2} \cdot \frac{e^{-\lambda_1}\lambda_1^{n}}{n!} + \frac{\beta_2^{(s)}\lambda_2}{\beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2} \cdot \frac{e^{-\lambda_2}\lambda_2^{n}}{n!}.
\end{aligned}$$

Notice, this is also a finite mixture distribution where both processes follow the Poisson distribution with λ_1, λ_2 respectively. However, we have updated values for $\beta_1^{(s)}$ and $\beta_2^{(s)}$. The weight update rule is

$$\beta_j^{(s+1)} = \frac{\beta_j^{(s)}\,\lambda_j}{\beta_1^{(s)}\lambda_1 + \beta_2^{(s)}\lambda_2}.$$

So, we have

$$P^{(s)}(S_N = k) = \frac{E[N^{(s)}]}{k} \sum_{i=1}^{k} i\,\alpha_i \cdot P^{(s+1)}(S_N = k - i).$$

Now we can begin the computation. Let $\lambda_1 = 3$, $\lambda_2 = 4$, $\beta_1^{(0)} = 0.6$, $\beta_2^{(0)} = 0.4$. We want to find $P^{(0)}(S_N = 4)$.

Computing the mixing weights at each level:

$$\begin{aligned}
&\beta_1^{(0)} = 0.6, \quad \beta_2^{(0)} = 0.4, \quad E[N^{(0)}] = (0.6)(3) + (0.4)(4) = 3.4.\\
&\beta_1^{(1)} = \frac{(0.6)(3)}{3.4} = \frac{1.8}{3.4}, \quad \beta_2^{(1)} = \frac{(0.4)(4)}{3.4} = \frac{1.6}{3.4}, \quad E[N^{(1)}] = \frac{1.8(3) + 1.6(4)}{3.4} = \frac{11.8}{3.4}.\\
&\beta_1^{(2)} = \frac{5.4}{11.8}, \quad \beta_2^{(2)} = \frac{6.4}{11.8}, \quad E[N^{(2)}] = \frac{5.4(3) + 6.4(4)}{11.8} = \frac{41.8}{11.8}.\\
&\beta_1^{(3)} = \frac{16.2}{41.8}, \quad \beta_2^{(3)} = \frac{25.6}{41.8}, \quad E[N^{(3)}] = \frac{16.2(3) + 25.6(4)}{41.8} = \frac{151}{41.8}.\\
&\beta_1^{(4)} = \frac{48.6}{151}, \quad \beta_2^{(4)} = \frac{102.4}{151}.
\end{aligned}$$

Base cases $P^{(s)}(S_N = 0) = \beta_1^{(s)} e^{-\lambda_1} + \beta_2^{(s)} e^{-\lambda_2}$:

$$P^{(1)}(S_N = 0) = \tfrac{1.8}{3.4}\, e^{-3} + \tfrac{1.6}{3.4}\, e^{-4}.$$

$$P^{(2)}(S_N = 0) = \tfrac{5.4}{11.8}\, e^{-3} + \tfrac{6.4}{11.8}\, e^{-4}.$$

$$P^{(3)}(S_N = 0) = \tfrac{16.2}{41.8}\, e^{-3} + \tfrac{25.6}{41.8}\, e^{-4}.$$

$$P^{(4)}(S_N = 0) = \tfrac{48.6}{151}\, e^{-3} + \tfrac{102.4}{151}\, e^{-4}.$$

Computing $P^{(s)}(S_N = 1)$:

$$\begin{aligned}
P^{(1)}(S_N = 1) &= \frac{E[N^{(1)}]}{1}\big[1 \cdot 0.05 \cdot P^{(2)}(S_N = 0)\big]\\
&= \frac{11.8}{3.4}\left[(0.05)\left(\frac{5.4}{11.8}\, e^{-3} + \frac{6.4}{11.8}\, e^{-4}\right)\right]\\
&= 0.07941176471\, e^{-3} + 0.09411764706\, e^{-4}.\\
P^{(2)}(S_N = 1) &= \frac{E[N^{(2)}]}{1}\big[0.05 \cdot P^{(3)}(S_N = 0)\big]\\
&= 0.06864406780\, e^{-3} + 0.10847457627\, e^{-4}.\\
P^{(3)}(S_N = 1) &= \frac{E[N^{(3)}]}{1}\big[0.05 \cdot P^{(4)}(S_N = 0)\big]\\
&= 0.05813397129\, e^{-3} + 0.12248803828\, e^{-4}.
\end{aligned}$$

Computing $P^{(s)}(S_N = 2)$:

$$P^{(1)}(S_N = 2) = \frac{E[N^{(1)}]}{2}\big[(0.05) \cdot P^{(2)}(S_N = 1) + (0.8) \cdot P^{(2)}(S_N = 0)\big]$$

$$= 0.64125\, e^{-3} + 0.76235294118\, e^{-4}.$$

$$\begin{aligned} P^{(2)}(S_N = 2) &= \frac{E[N^{(2)}]}{2}\big[(0.05)\cdot P^{(3)}(S_N = 1) + (0.8)\cdot P^{(3)}(S_N = 0)\big] \\ &= 0.55430084746\, e^{-3} + 0.87864406780\, e^{-4}. \end{aligned}$$

Computing $P^{(1)}(S_N = 3)$:

$$\begin{aligned} P^{(1)}(S_N = 3) &= \frac{E[N^{(1)}]}{3}\big[(0.05)\cdot P^{(2)}(S_N = 2) + (0.8)\cdot P^{(2)}(S_N = 1) + (0.3)\cdot P^{(2)}(S_N = 0)\big] \\ &= 0.25441544118\, e^{-3} + 0.33945098039\, e^{-4}. \end{aligned}$$

Finally, $P^{(0)}(S_N = 4)$:

$$\begin{aligned} P^{(0)}(S_N = 4) &= \frac{E[N^{(0)}]}{4}\big[(0.05)\cdot P^{(1)}(S_N = 3) + (0.8)\cdot P^{(1)}(S_N = 2) \\ &\qquad + (0.3)\cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\ &= \frac{3.4}{4}\big[(0.05)(0.25441544118\, e^{-3} + 0.33945098039\, e^{-4}) \\ &\qquad + (0.8)(0.64125\, e^{-3} + 0.76235294118\, e^{-4}) \\ &\qquad + (0.3)(0.07941176471\, e^{-3} + 0.09411764706\, e^{-4}) \\ &\qquad + \big(\tfrac{1.8}{3.4}\, e^{-3} + \tfrac{1.6}{3.4}\, e^{-4}\big)\big] \\ &= \boxed{0.91711265625\, e^{-3} + 0.95682666667\, e^{-4}}. \end{aligned}$$

6. Finite Poisson Mixture Generalization

Next, we will describe a more general case of the Finite Mixture of Poisson Distributions. Consider a discrete mixture distribution where the event is generated by one or more random processes. The event is generated by one of t processes, which follows a Poisson distribution with mean λ_i for $i \in \mathbb{Z}$ with $0 < i \leq t$. The probability of process i is β_i for $i \in \mathbb{Z}$ with $0 < i \leq t$ and where $\beta_1 + \beta_2 + \cdots + \beta_t = 1$.

Now we are ready to define our probability mass function. Let $s \in \mathbb{Z}$ with $0 \leq s$, and let s represent the level of recursion. Let the parameters of this distribution be $s, \lambda_1, \ldots, \lambda_t$.

$$P^{(s)}(N = n) = \beta_1^{(s)} \cdot \frac{e^{-\lambda_1}\lambda_1^n}{n!} + \beta_2^{(s)} \cdot \frac{e^{-\lambda_2}\lambda_2^n}{n!} + \cdots + \beta_t^{(s)} \cdot \frac{e^{-\lambda_t}\lambda_t^n}{n!}, \qquad n \geq 0.$$

Corollary 2 (Recursive Properties of the Generalized Finite Mixture of Poisson Distributions)**.** *Let $N^{(s)}$ be a mixed Poisson random variable at recursion level s with probability mass function*

$$P^{(s)}(N = n) = \sum_{j=1}^{t} \left(\frac{\beta_j \, \lambda_j^s}{\sum_{i=1}^t \beta_i \, \lambda_i^s} \right) \cdot \frac{e^{-\lambda_j}\lambda_j^n}{n!},$$

where $\sum_{i=1}^t \beta_i = 1$, $\beta_i > 0$, and $\lambda_i > 0$. Let $M^{(s)}$ be the random variable satisfying the compound identity condition associated with $N^{(s)}$. We claim that for all $s \geq 0$:

(1) $E[N^{(s)}] = \dfrac{\sum_{i=1}^t \beta_i \, \lambda_i^{s+1}}{\sum_{j=1}^t \beta_j \, \lambda_j^s}$.

(2) *The distribution of $M^{(s)} - 1$ is identical to the distribution of $N^{(s+1)}$.*

Proof (by induction). Base Case $(s = 0)$: For $s = 0$, the weights are β_i. First, we compute the expected value of $E[N^{(0)}]$:

$$E[N^{(0)}] = \sum_{n=0}^{\infty} n \left[\sum_{i=1}^t \beta_i \cdot \frac{e^{-\lambda_i}\lambda_i^n}{n!} \right] = \sum_{i=1}^t \beta_i \, \lambda_i.$$

This matches the form of Claim 1 where the denominator $\sum_{i=1}^t \beta_i \, \lambda_i^0 = \sum_{i=1}^t \beta_i = 1$.

Next, we apply the compound random identity and see that,

$$\begin{aligned}
P(M^{(0)} - 1 = n) = P(M^{(0)} = n + 1) &= \frac{(n+1)\, P^{(0)}(N = n+1)}{E[N^{(0)}]} \\
&= \frac{(n+1)\left[\beta_1 \cdot \dfrac{e^{-\lambda_1}\lambda_1^{n+1}}{(n+1)!} + \cdots + \beta_t \cdot \dfrac{e^{-\lambda_t}\lambda_t^{n+1}}{(n+1)!}\right]}{\beta_1\lambda_1 + \cdots + \beta_t\lambda_t} \\
&= \frac{n+1}{\sum_{j=1}^t \beta_j\lambda_j} \cdot \sum_{i=1}^t \beta_i \cdot \frac{e^{-\lambda_i}\lambda_i^{n+1}}{(n+1)!}
\end{aligned}$$

$$= \sum_{i=1}^{t} \frac{\beta_i \, \lambda_i}{\sum_{j=1}^{t} \beta_j \, \lambda_j} \cdot \frac{e^{-\lambda_i} \lambda_i^n}{n!}.$$

This is exactly the definition of $P^{(1)}(N = n)$. Thus, the claim holds for $s = 0$.

Inductive Step: Assume the statement is true for $s = k$. That is, $M^{(k)} - 1$ is equivalent to $N^{(k+1)}$ and $E[N^{(k)}]$ follows the claim. We need to show the statement is true for $s = k+1$. We must determine the distribution of $M^{(k+1)} - 1$.

We have,

$$P(M^{(k+1)} - 1 = n) = P(M^{(k+1)} = n + 1) = \frac{(n+1) \, P^{(k+1)}(N = n+1)}{E[N^{(k+1)}]}.$$

First, we compute $E[N^{(k+1)}]$:

$$\begin{aligned} E[N^{(k+1)}] &= \sum_{n=0}^{\infty} n \cdot \sum_{i=1}^{t} \left(\frac{\beta_i \, \lambda_i^{k+1}}{\sum_{j=1}^{t} \beta_j \, \lambda_j^{k+1}} \right) \cdot \frac{e^{-\lambda_i} \lambda_i^n}{n!} \\ &= \sum_{i=1}^{t} \left(\frac{\beta_i \, \lambda_i^{k+1}}{\sum_{j=1}^{t} \beta_j \, \lambda_j^{k+1}} \right) \cdot \lambda_i = \frac{\sum_{i=1}^{t} \beta_i \, \lambda_i^{k+2}}{\sum_{j=1}^{t} \beta_j \, \lambda_j^{k+1}}. \end{aligned}$$

This confirms Claim 1 for $s = k+1$. Next, we substitute this into the probability expression:

$$P(M^{(k+1)} - 1 = n) = \frac{n+1}{\frac{\sum_{i=1}^{t} \beta_i \lambda_i^{k+2}}{\sum_{j=1}^{t} \beta_j \lambda_j^{k+1}}} \sum_{i=1}^{t} \left(\frac{\beta_i \lambda_i^{k+1}}{\sum_{j=1}^{t} \beta_j \lambda_j^{k+1}} \right) \cdot \frac{e^{-\lambda_i} \lambda_i^{n+1}}{(n+1)!}.$$

We cancel the common denominator terms $\sum_{j=1}^{t} \beta_j \lambda_j^{k+1}$ and we have

$$\begin{aligned} &= \frac{n+1}{\sum_{j=1}^{t} \beta_j \lambda_j^{k+2}} \sum_{i=1}^{t} \beta_i \lambda_i^{k+1} \cdot \frac{e^{-\lambda_i} \lambda_i^{n+1}}{(n+1)!} \\ &= \sum_{i=1}^{t} \frac{\beta_i \lambda_i^{k+2}}{\sum_{j=1}^{t} \beta_j \lambda_j^{k+2}} \cdot \frac{e^{-\lambda_i} \lambda_i^{n}}{n!}. \end{aligned}$$

This expression is exactly the PMF for $N^{(k+2)}$ as desired.

So, by the principle of mathematical induction, for all $s \in \mathbb{N}$, the compound identity variable associated with $N^{(s)}$ shifts the distribution to $N^{(s+1)}$, and the expected value satisfies the derived closed form. $\square$

Now we are ready to construct our new recursive expression by applying Corollary 1. From

$$P(S_N = k) = \frac{1}{k}\, E[N] \sum_{i=1}^{k} i\, \alpha_i\, P(S_{M-1} = k - i),$$

we arrive at the updated recursive equation

$$P^{(s)}(S_N = k) = \frac{1}{k} \cdot \frac{\sum_{i=1}^{t} \beta_i\, \lambda_i^{s+1}}{\sum_{i=1}^{t} \beta_i\, \lambda_i^{s}} \cdot \sum_{i=1}^{k} i\, \alpha_i \cdot P^{(s+1)}(S_N = k - i).$$

Example 5 (Revisiting Example 1)**.** Now we can compute the same compound random variable from Example 1, but with our updated recursive equation. Like before, let N be a random variable of the Poisson distribution with parameter $\lambda_1 = 3$. So the probability mass function for the distribution of N is

$$P^{(s)}(N = n) = \beta_1 \cdot \frac{e^{-\lambda}\, \lambda_1^{n}}{n!}, \quad \text{where } \beta_1 = 1.$$

Furthermore,

$$E[N^{(s)}] = \frac{\sum_{i=1}^{1} \beta_i\, \lambda_i^{s+1}}{\sum_{i=1}^{1} \beta_i\, \lambda_i^{s}} = \frac{\beta_1\, \lambda_1^{s+1}}{\beta_1\, \lambda_1^{s}} = \frac{\lambda_1^{s+1}}{\lambda_1^{s}} = \lambda_1 = 3, \qquad \forall\, s \in \mathbb{Z},\ s \geq 0.$$

Now we calculate $P^{(0)}(S_N = 4)$. We want to find

$$P^{(0)}(S_N = 4) = \frac{3}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + 4(0.25) \cdot P^{(1)}(S_N = 0)\big].$$

We can solve each of these recursive expressions for all $s \in \mathbb{Z}$, $s \geq 0$. So we have,

$$P^{(s)}(S_N = 0) = P(N^{(s)} = 0) = 1 \cdot \frac{e^{-\lambda}\, \lambda^0}{0!} = e^{-\lambda} = e^{-3}, \qquad \forall\, s \geq 0.$$

$$P^{(s)}(S_N = 1) = \frac{3}{1}\big[1 \cdot (0.05) \cdot P(S_N = 0)\big] = 3(0.05)\, e^{-3} = 0.15\, e^{-3}, \qquad \forall\, s \geq 0.$$

$$\begin{aligned} P^{(s)}(S_N = 2) &= \frac{3}{2}\big[(0.05) \cdot P(S_N = 1) + 2(0.4) \cdot P(S_N = 0)\big] \\ &= \frac{3}{2}\big[(0.05)(0.15\, e^{-3}) + (0.8)\, e^{-3}\big] = 1.21125\, e^{-3}, \qquad \forall\, s \geq 0. \end{aligned}$$

$$\begin{aligned} P^{(s)}(S_N = 3) &= \frac{3}{3}\big[(0.05) \cdot P(S_N = 2) + (0.8) \cdot P(S_N = 1) + 3(0.1) \cdot P(S_N = 0)\big] \\ &= (0.05)(1.21125\, e^{-3}) + (0.8)(0.15\, e^{-3}) + (0.3)\, e^{-3} = 0.4805625\, e^{-3}, \qquad \forall\, s \geq 0. \end{aligned}$$

Now we have everything we need to compute $P^{(0)}(S_N = 4)$. We have,

$$\begin{aligned} P^{(0)}(S_N = 4) &= \frac{3}{4}\left[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + 4(0.25) \cdot P^{(1)}(S_N = 0)\right] \\ &= \frac{3}{4}\left[0.05(0.4805625\, e^{-3}) + 0.8(1.21125\, e^{-3}) + 0.3(0.15\, e^{-3}) + e^{-3}\right] \\ &= 1.528521094\, e^{-3}. \end{aligned}$$

Notice, this is the same as our result from Example 1.

Example 6 (Revisiting Example 4)**.** We will consider again the case of a discrete mixture distribution where the event is generated by one of two random processes. The event is either generated by process one, which follows a Poisson distribution with mean λ_1, or process two, which follows a Poisson distribution with mean λ_2. Like in Example 4, we let $\lambda_1 = 3$, $\lambda_2 = 4$, $\beta_1^{(0)} = 0.6$, $\beta_2^{(0)} = 0.4$. We want to find $P^{(0)}(S_N = 4)$.

So we have,

$$P^{(s)}(S_N = k) = \frac{1}{k}\left[\frac{(0.6)(3)^{s+1} + (0.4)(4)^{s+1}}{(0.6)(3)^s + (0.4)(4)^s}\right] \sum_{i=1}^{k} i\, \alpha_i \cdot P^{(s+1)}(S_N = k - i).$$

$$P^{(0)}(S_N = 4) = \frac{1}{4}\left[\frac{(0.6)(3)^1 + (0.4)(4)^1}{(0.6)(3)^0 + (0.4)(4)^0}\right] \sum_{i=1}^{4} i\, \alpha_i \cdot P^{(1)}(S_N = 4 - i).$$

Base cases:

$$\begin{aligned} P^{(1)}(S_N = 0) &= \frac{(0.6)(3)^1}{(0.6)(3)^1 + (0.4)(4)^1}\, e^{-3} + \frac{(0.4)(4)^1}{(0.6)(3)^1 + (0.4)(4)^1}\, e^{-4} = \frac{1.8}{3.4}\, e^{-3} + \frac{1.6}{3.4}\, e^{-4}. \\ P^{(2)}(S_N = 0) &= \frac{(0.6)(3)^2}{(0.6)(3)^2 + (0.4)(4)^2}\, e^{-3} + \frac{(0.4)(4)^2}{(0.6)(3)^2 + (0.4)(4)^2}\, e^{-4} = \frac{5.4}{11.8}\, e^{-3} + \frac{6.4}{11.8}\, e^{-4}. \\ P^{(3)}(S_N = 0) &= \frac{16.2}{41.8}\, e^{-3} + \frac{25.6}{41.8}\, e^{-4}. \\ P^{(4)}(S_N = 0) &= \frac{48.6}{151}\, e^{-3} + \frac{102.4}{151}\, e^{-4}. \end{aligned}$$

Computing $P^{(s)}(S_N = 1)$:

$$
\begin{aligned}
P^{(1)}(S_N = 1) &= \left[\frac{(0.6)(3)^2 + (0.4)(4)^2}{(0.6)(3)^1 + (0.4)(4)^1}\right]\left[1 \cdot 0.05 \cdot P^{(2)}(S_N = 0)\right] \\
&= \frac{11.8}{3.4}\left[0.05\left(\frac{5.4}{11.8}\, e^{-3} + \frac{6.4}{11.8}\, e^{-4}\right)\right] = 0.07941176471\, e^{-3} + 0.09411764706\, e^{-4}. \\
P^{(2)}(S_N = 1) &= \frac{41.8}{11.8}\left[0.05 \cdot P^{(3)}(S_N = 0)\right] = 0.06864406780\, e^{-3} + 0.10847457627\, e^{-4}. \\
P^{(3)}(S_N = 1) &= \frac{151}{41.8}\left[0.05 \cdot P^{(4)}(S_N = 0)\right] = 0.05813397129\, e^{-3} + 0.12248803828\, e^{-4}.
\end{aligned}
$$

Computing $P^{(s)}(S_N = 2)$:

$$
\begin{aligned}
P^{(1)}(S_N = 2) &= \frac{1}{2} \cdot \frac{11.8}{3.4}\left[(0.05) \cdot P^{(2)}(S_N = 1) + (0.8) \cdot P^{(2)}(S_N = 0)\right] \\
&= 0.64125\, e^{-3} + 0.76235294118\, e^{-4}. \\
P^{(2)}(S_N = 2) &= \frac{1}{2} \cdot \frac{41.8}{11.8}\left[(0.05) \cdot P^{(3)}(S_N = 1) + (0.8) \cdot P^{(3)}(S_N = 0)\right] \\
&= 0.55430084746\, e^{-3} + 0.87864406780\, e^{-4}.
\end{aligned}
$$

Computing $P^{(1)}(S_N = 3)$:

$$
\begin{aligned}
P^{(1)}(S_N = 3) &= \frac{1}{3} \cdot \frac{11.8}{3.4}\left[(0.05) \cdot P^{(2)}(S_N = 2) + (0.8) \cdot P^{(2)}(S_N = 1) + (0.3) \cdot P^{(2)}(S_N = 0)\right] \\
&= 0.25441544118\, e^{-3} + 0.33945098039\, e^{-4}.
\end{aligned}
$$

Finally, $P^{(0)}(S_N = 4)$:

$$
\begin{aligned}
P^{(0)}(S_N = 4) = \frac{1}{4} \cdot \frac{3.4}{1}\big[&(0.05) \cdot P^{(1)}(S_N = 3) + (0.8) \cdot P^{(1)}(S_N = 2) \\
&+ (0.3) \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\
= {} & \boxed{0.91711265625\, e^{-3} + 0.95682666667\, e^{-4}}.
\end{aligned}
$$

7. Generalized Computation with a Finite Mixture of Distributions

We have shown in Example 1 with N in the Poisson distribution that

$$P^{(s+1)}(M^{(s)} - 1 = n) = P^{(s)}(N = n).$$

We have shown in Example 2 with N in the negative binomial distribution that

$$P^{(s)}(M^{(s)} - 1 = n) = P^{(s+1)}(N = n),$$

where the negative binomial parameter shifts from $r + s$ to $r + s + 1$. Similarly, we have shown in Example 3 with N in the binomial distribution that

$$P^{(s)}(M^{(s)} - 1 = n) = P^{(s+1)}(N = n),$$

where the binomial parameter shifts from $r - s$ to $r - s - 1$.

Notice that for each of these distributions, when calculating $P(M^{(s)} - 1 = n)$, at a given level of recursion, we arrive at either $P^{(s)}(N = n)$ or a version of $P^{(s)}(N = n)$ with a parameter value change that is a function of the level of recursion.

We will now show that by moving the level of recursion to a parameter of the probability mass function of the finite mixture of distributions we can compute a compound random variable with the counting distribution in any finite mixture of any of these three distributions.

Corollary 3 (Recursive Properties of the Finite Mixture Distributions)**.** *Let $s \in \mathbb{Z}$ with $0 \leq s$, and let s represent the level of recursion. Let $N^{(s)}$ be a mixed random variable of Poisson, Binomial, or Negative Binomial distribution at recursion level s with probability mass function:*

$$P^{(s)}(N = n) = \sum_{j=1}^{t} \beta_j^{(s)} \, P_{r_j}(N^{(s)} = n), \qquad n \geq 0,$$

where $\sum_{j=1}^{t} \beta_j^{(s)} = 1$. Let $M^{(s)}$ be the random variable satisfying the compound identity condition associated with $N^{(s)}$. We claim that for all $s \geq 0$:

(1) $E[N^{(s)}] = \sum_{j=1}^{t} \beta_j^{(s)} \, E[N_j^{(s)}]$.

(2) *The distribution of $M^{(s)} - 1$ is identical to the distribution of $N^{(s+1)}$, where the new weights are updated such that*

$$\beta_j^{(s+1)} = \beta_j^{(s)} \cdot \frac{E[N_j^{(s)}]}{E[N^{(s)}]}.$$

Proof (by induction). Base Case ($s = 0$): For $s = 0$, the weights are $\beta_j^{(0)}$. First, we compute the expected value of $E[N^{(0)}]$:

$$E[N^{(0)}] = \sum_{n=0}^{\infty} n \cdot \left[\sum_{j=1}^{t} \beta_j^{(0)} P_{r_j}(n)\right] = \sum_{j=1}^{t} \beta_j^{(0)} E[N_j^{(0)}].$$

Thus, Claim 1 holds for case $s = 0$.

Next, we apply the compound random identity and see that,

$$\begin{aligned} P(M^{(0)} - 1 = n) = P(M^{(0)} = n+1) &= \frac{(n+1)\, P(N^{(0)} = n+1)}{E[N^{(0)}]} \\ &= \frac{1}{E[N^{(0)}]} \sum_{j=1}^{t} \beta_j^{(0)} \Big[(n+1) \cdot P_{r_j}(N^{(0)} = n+1)\Big]. \end{aligned}$$

For each component distribution, the compound identity shifts the parameter by one level of recursion. For the negative binomial distribution,

$$(n+1)\, P_r(N = n+1) = E[N]\, P_{r+1}(N = n);$$

for the binomial distribution,

$$(n+1)\, P_r(N = n+1) = E[N]\, P_{r-1}(N = n);$$

and for the Poisson distribution the parameter is unchanged,

$$(n+1)\, P_\lambda(N = n+1) = E[N]\, P_\lambda(N = n).$$

Substituting the appropriate identity for each component, we have

$$\begin{aligned} P(M^{(0)} - 1 = n) &= \frac{1}{E[N^{(0)}]} \sum_{j=1}^{t} \beta_j^{(0)} E[N_j^{(0)}] \cdot P_{r_j}(N^{(1)} = n) \\ &= \sum_{j=1}^{t} \frac{\beta_j^{(0)} E[N_j^{(0)}]}{E[N^{(0)}]} \cdot P_{r_j}(N^{(1)} = n), \end{aligned}$$

where $P_{r_j}(N^{(1)} = n)$ denotes the jth component advanced to level 1 (parameter $r_j + 1$ for negative binomial, $r_j - 1$ for binomial, and unchanged for Poisson). Defining $\beta_j^{(1)} = \frac{\beta_j^{(0)} E[N_j^{(0)}]}{E[N^{(0)}]}$, this is exactly the mixture $P^{(1)}(N = n)$, and the new weights sum to one since $\sum_{j=1}^{t} \beta_j^{(1)} = \frac{1}{E[N^{(0)}]} \sum_{j=1}^{t} \beta_j^{(0)} E[N_j^{(0)}] = 1$. Thus, the claim holds for $s = 0$.

Inductive Step: Assume the statement is true for $s = k$. That is, $M^{(k)} - 1$ is equivalent to $N^{(k+1)}$ and $E[N^{(k)}]$ follows the claim. We need to show the statement is true for $s = k+1$. By the inductive hypothesis, $N^{(k+1)}$ is the mixture $\sum_{j=1}^{t} \beta_j^{(k+1)} P_{r_j}(N^{(k+1)} = n)$, so by linearity of expectation,

$$E[N^{(k+1)}] = \sum_{n=0}^{\infty} n \sum_{j=1}^{t} \beta_j^{(k+1)} P_{r_j}(N^{(k+1)} = n) = \sum_{j=1}^{t} \beta_j^{(k+1)} E[N_j^{(k+1)}],$$

which confirms Claim 1 for $s = k+1$. Next, we determine the distribution of $M^{(k+1)} - 1$. We have,

$$\begin{aligned} P(M^{(k+1)} - 1 = n) = P(M^{(k+1)} = n+1) &= \frac{(n+1)\, P(N^{(k+1)} = n+1)}{E[N^{(k+1)}]} \\ &= \frac{1}{E[N^{(k+1)}]} \sum_{j=1}^{t} \beta_j^{(k+1)} \Big[(n+1) \cdot P_{r_j}(N^{(k+1)} = n+1)\Big]. \end{aligned}$$

Applying the same per-component shift identity used in the base case, now advancing each component from level $k+1$ to level $k+2$, we have

$$\begin{aligned} P(M^{(k+1)} - 1 = n) &= \frac{1}{E[N^{(k+1)}]} \sum_{j=1}^{t} \beta_j^{(k+1)} E[N_j^{(k+1)}] \cdot P_{r_j}(N^{(k+2)} = n) \\ &= \sum_{j=1}^{t} \frac{\beta_j^{(k+1)} E[N_j^{(k+1)}]}{E[N^{(k+1)}]} \cdot P_{r_j}(N^{(k+2)} = n). \end{aligned}$$

This expression is exactly the PMF for $N^{(k+2)}$ with updated weights $\beta_j^{(k+2)} = \dfrac{\beta_j^{(k+1)} E[N_j^{(k+1)}]}{E[N^{(k+1)}]}$, as desired.

So, by the principle of mathematical induction, for all $s \in \mathbb{N}$, the compound identity variable associated with $N^{(s)}$ shifts the distribution to $N^{(s+1)}$, and the expected value satisfies the derived closed form. $\square$

Now we are ready to construct our new recursive expression by applying Corollary 1. From

$$P(S_N = k) = \frac{1}{k} E[N] \sum_{i=1}^{k} i\, \alpha_i\, P(S_{M-1} = k - i),$$

we arrive at the updated recursive equation

$$P^{(s)}(S_N = k) = \frac{E[N^{(s)}]}{k} \cdot \sum_{i=1}^{k} i\,\alpha_i \cdot P^{(s+1)}(S_N = k - i),$$

where $E[N^{(s)}] = \sum_{j=1}^{t} \beta_j^{(s)}\, E[N_j^{(s)}]$ and the computation of $P^{(s+1)}$ uses $\beta^{(s+1)}$ and parameters appropriate to the distribution of P_i, that is, $(r - s)$ for the binomial distribution, $(r + s)$ for the negative binomial distribution and no parameter change for the Poisson distribution.

Example 7 (N in a Finite Mixture of Binomial Distributions)**.** Consider a discrete mixture distribution where the event is generated by one of two random processes. The event is either generated by process one, which follows a Binomial distribution with $r_1 = 8$ and $p_1 = 0.5$, or process two, which follows a Binomial distribution with $r_2 = 10$ and $p_2 = 0.7$. The probability of process one is $\beta_1^{(0)} = 0.6$. The probability of process two is $\beta_2^{(0)} = 0.4$. Notice that $\beta_1^{(0)} + \beta_2^{(0)} = 1$. We want to find $P(S_N = 4)$.

So, the probability mass function for this mixed distribution is

$$P^{(s)}(N = n) = \sum_{j=1}^{2} \beta_j^{(s)} \binom{r_j - s}{n} p_j^n (1 - p_j)^{(r_j - s) - n}, \quad 0 \le n \le \max(r_1 - s,\, r_2 - s).$$

Computing the mixing weights at each level:

$$E[N^{(0)}] = \beta_1^{(0)}(r_1 - 0)p_1 + \beta_2^{(0)}(r_2 - 0)p_2 = 0.6 \cdot 8 \cdot 0.5 + 0.4 \cdot 10 \cdot 0.7 = 5.2.$$

$$\beta_1^{(1)} = \beta_1^{(0)} \cdot \frac{E[N_1^{(0)}]}{E[N^{(0)}]} = \frac{2.4}{5.2} = 0.46153846154.$$

$$\beta_2^{(1)} = \beta_2^{(0)} \cdot \frac{E[N_2^{(0)}]}{E[N^{(0)}]} = \frac{2.8}{5.2} = 0.53846153846.$$

$$E[N^{(1)}] = \beta_1^{(1)}(r_1 - 1)p_1 + \beta_2^{(1)}(r_2 - 1)p_2$$

$$= 0.46153846154 \cdot 7 \cdot 0.5 + 0.53846153846 \cdot 9 \cdot 0.7 = 5.00769230769.$$

$$\beta_1^{(2)} = \beta_1^{(1)} \cdot \frac{E[N_1^{(1)}]}{E[N^{(1)}]} = \frac{1.61538461538}{5.00769230769} = 0.32258064516.$$

$$\beta_2^{(2)} = \beta_2^{(1)} \cdot \frac{E[N_2^{(1)}]}{E[N^{(1)}]} = \frac{3.39230769231}{5.00769230769} = 0.67741935484.$$

$$E[N^{(2)}] = \beta_1^{(2)}(r_1-2)p_1 + \beta_2^{(2)}(r_2-2)p_2$$
$$= 0.32258064516 \cdot 6 \cdot 0.5 + 0.67741935484 \cdot 8 \cdot 0.7 = 4.76129032258.$$
$$\beta_1^{(3)} = \beta_1^{(2)} \cdot \frac{E[N_1^{(2)}]}{E[N^{(2)}]} = \frac{0.96774193548}{4.76129032258} = 0.20325203252.$$
$$\beta_2^{(3)} = \beta_2^{(2)} \cdot \frac{E[N_2^{(2)}]}{E[N^{(2)}]} = \frac{3.79354838710}{4.76129032258} = 0.79674796748.$$
$$E[N^{(3)}] = \beta_1^{(3)}(r_1-3)p_1 + \beta_2^{(3)}(r_2-3)p_2$$
$$= 0.20325203252 \cdot 5 \cdot 0.5 + 0.79674796748 \cdot 7 \cdot 0.7 = 4.41219512195.$$
$$\beta_1^{(4)} = \beta_1^{(3)} \cdot \frac{E[N_1^{(3)}]}{E[N^{(3)}]} = \frac{0.50813008130}{4.41219512195} = 0.11516491616.$$
$$\beta_2^{(4)} = \beta_2^{(3)} \cdot \frac{E[N_2^{(3)}]}{E[N^{(3)}]} = \frac{3.90406504065}{4.41219512195} = 0.88483508384.$$
$$E[N^{(4)}] = \beta_1^{(4)}(r_1-4)p_1 + \beta_2^{(4)}(r_2-4)p_2$$
$$= 0.11516491616 \cdot 4 \cdot 0.5 + 0.88483508384 \cdot 6 \cdot 0.7 = 3.94663718445.$$

Now we can begin the computation to find $P^{(0)}(S_N = 4)$. We want to find,

$$P^{(0)}(S_N=4) = \frac{E[N^{(0)}]}{4}\big[0.05 \cdot P^{(1)}(S_N=3) + 0.8 \cdot P^{(1)}(S_N=2) + 0.3 \cdot P^{(1)}(S_N=1) + P^{(1)}(S_N=0)\big].$$

Base cases:

$$P^{(s)}(S_N=0) = \beta_1^{(s)} \cdot (1-p_1)^{r_1-s} + \beta_2^{(s)} \cdot (1-p_2)^{r_2-s} = \beta_1^{(s)} \cdot (0.5)^{8-s} + \beta_2^{(s)} \cdot (0.3)^{10-s}.$$

$$P^{(1)}(S_N=0) = 0.46153846154\,(0.5)^7 + 0.53846153846\,(0.3)^9 = 0.00361636777.$$
$$P^{(2)}(S_N=0) = 0.32258064516\,(0.5)^6 + 0.67741935484\,(0.3)^8 = 0.00508476806.$$
$$P^{(3)}(S_N=0) = 0.20325203252\,(0.5)^5 + 0.79674796748\,(0.3)^7 = 0.00652587480.$$
$$P^{(4)}(S_N=0) = 0.11516491616\,(0.5)^4 + 0.88483508384\,(0.3)^6 = 0.00784285204.$$

Computing $P^{(s)}(S_N = 1)$:

$$P^{(1)}(S_N=1) = \frac{E[N^{(1)}]}{1}\big[0.05 \cdot P^{(2)}(S_N=0)\big] = 5.00769230769(0.05)(0.00508476806) = 0.00127314770.$$
$$P^{(2)}(S_N=1) = \frac{E[N^{(2)}]}{1}\big[0.05 \cdot P^{(3)}(S_N=0)\big] = 4.76129032258(0.05)(0.00652587480) = 0.00155357923.$$

$$P^{(3)}(S_N = 1) = \frac{E[N^{(3)}]}{1}\big[0.05 \cdot P^{(4)}(S_N = 0)\big] = 4.41219512195(0.05)(0.00784285204) = 0.00173020967.$$

Computing $P^{(s)}(S_N = 2)$:

$$\begin{aligned}
P^{(1)}(S_N = 2) &= \frac{E[N^{(1)}]}{2}\big[0.05 \cdot P^{(2)}(S_N = 1) + 0.8 \cdot P^{(2)}(S_N = 0)\big] \\
&= \frac{5.00769230769}{2}\big[0.05(0.00155357923) + 0.8(0.00508476806)\big] = 0.01037967774. \\
P^{(2)}(S_N = 2) &= \frac{E[N^{(2)}]}{2}\big[0.05 \cdot P^{(3)}(S_N = 1) + 0.8 \cdot P^{(3)}(S_N = 0)\big] \\
&= \frac{4.76129032258}{2}\big[0.05(0.00173020967) + 0.8(0.00652587480)\big] = 0.01263458457.
\end{aligned}$$

Computing $P^{(1)}(S_N = 3)$:

$$\begin{aligned}
P^{(1)}(S_N = 3) &= \frac{E[N^{(1)}]}{3}\big[0.05 \cdot P^{(2)}(S_N = 2) + 0.8 \cdot P^{(2)}(S_N = 1) + 0.3 \cdot P^{(2)}(S_N = 0)\big] \\
&= \frac{5.00769230769}{3}\big[0.05(0.01263458457) + 0.8(0.00155357923) + 0.3(0.00508476806)\big] \\
&= 0.00567542306.
\end{aligned}$$

Computing $P^{(0)}(S_N = 4)$:

$$\begin{aligned}
P^{(0)}(S_N = 4) &= \frac{E[N^{(0)}]}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\
&= \frac{5.2}{4}\big[0.05(0.00567542306) + 0.8(0.01037967774) \\
&\qquad + 0.3(0.00127314770) + 0.00361636777\big] \\
&= \boxed{0.01636157305}.
\end{aligned}$$

Example 8 (N in a Finite Mixture of Negative Binomial Distributions)**.** Consider a discrete mixture distribution where the event is generated by one of two random processes. The event is either generated by process one, which follows a Negative Binomial distribution with $r_1 = 4$ and $p_1 = 0.6$, or process two, which follows a Negative Binomial distribution with $r_2 = 6$ and $p_2 = 0.5$. The probability of process one is $\beta_1^{(0)} = 0.5$. The probability of process two is $\beta_2^{(0)} = 0.5$. Notice that $\beta_1^{(0)} + \beta_2^{(0)} = 1$. We want to find $P(S_N = 4)$.

So, the probability mass function for this mixed distribution is

$$P^{(s)}(N = n) = \sum_{j=1}^{2} \beta_j^{(s)} \binom{n + r_j + s - 1}{n} p_j^{r_j+s}(1 - p_j)^n, \quad 0 \le n \le \max(r_1 + s,\, r_2 + s).$$

Computing the mixing weights at each level:

$$\begin{aligned}
E[N^{(0)}] &= 0.5\left(\frac{(4+0)(0.4)}{0.6}\right) + 0.5\left(\frac{(6+0)(0.5)}{0.5}\right) = 4.33333333333.\\
\beta_1^{(1)} &= 0.5 \cdot \frac{1.33333333333}{4.33333333333} = 0.30769230769.\\
\beta_2^{(1)} &= 0.5 \cdot \frac{3}{4.33333333333} = 0.69230769231.\\
E[N^{(1)}] &= 0.30769230769 \cdot \frac{5 \cdot 0.4}{0.6} + 0.69230769231 \cdot \frac{7 \cdot 0.5}{0.5} = 5.87179487179.\\
\beta_1^{(2)} &= 0.30769230769 \cdot \frac{1.02564102564}{5.87179487179} = 0.17467248908.\\
\beta_2^{(2)} &= 0.69230769231 \cdot \frac{4.84615384615}{5.87179487179} = 0.82532751092.\\
E[N^{(2)}] &= 0.17467248908 \cdot \frac{6 \cdot 0.4}{0.6} + 0.82532751092 \cdot \frac{8 \cdot 0.5}{0.5} = 7.30131004367.\\
\beta_1^{(3)} &= 0.17467248908 \cdot \frac{0.69868995633}{7.30131004367} = 0.09569377990.\\
\beta_2^{(3)} &= 0.82532751092 \cdot \frac{6.60262008734}{7.30131004367} = 0.90430622010.\\
E[N^{(3)}] &= 0.09569377990 \cdot \frac{7 \cdot 0.4}{0.6} + 0.90430622010 \cdot \frac{9 \cdot 0.5}{0.5} = 8.58532695375.\\
\beta_1^{(4)} &= 0.09569377990 \cdot \frac{0.44664430295}{8.58532695375} = 0.05201560468.\\
\beta_2^{(4)} &= 0.90430622010 \cdot \frac{8.13875598086}{8.58532695375} = 0.94798439532.\\
E[N^{(4)}] &= 0.05201560468 \cdot \frac{8 \cdot 0.4}{0.6} + 0.94798439532 \cdot \frac{10 \cdot 0.5}{0.5} = 9.75726051149.
\end{aligned}$$

Now we can begin the computation to find $P^{(0)}(S_N = 4)$. We want to find,

$$P^{(0)}(S_N = 4) = \frac{E[N^{(0)}]}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big].$$

Base cases $P^{(s)}(S_N = 0) = \beta_1^{(s)}\, p_1^{r_1+s} + \beta_2^{(s)}\, p_2^{r_2+s}$:

$$P^{(0)}(S_N = 0) = 0.5(0.6)^4 + 0.5(0.5)^6 = 0.0726125.$$

$$P^{(1)}(S_N = 0) = 0.30769230769(0.6)^5 + 0.69230769231(0.5)^7 = 0.02933480769.$$

$$P^{(2)}(S_N = 0) = 0.17467248908(0.6)^6 + 0.82532751092(0.5)^8 = 0.01137345524.$$

$$P^{(3)}(S_N = 0) = 0.09569377990(0.6)^7 + 0.90430622010(0.5)^9 = 0.00444503648.$$

$$P^{(4)}(S_N = 0) = 0.05201560468(0.6)^8 + 0.94798439532(0.5)^{10} = 0.00179942843.$$

Computing $P^{(s)}(S_N = 1)$:

$$P^{(1)}(S_N = 1) = \frac{E[N^{(1)}]}{1}\big[0.05 \cdot P^{(2)}(S_N = 0)\big] = 5.87179487179(0.05)(0.01137345524) = 0.00333912981.$$

$$P^{(2)}(S_N = 1) = \frac{E[N^{(2)}]}{1}\big[0.05 \cdot P^{(3)}(S_N = 0)\big] = 7.30131004367(0.05)(0.00444503648) = 0.00162272948.$$

$$P^{(3)}(S_N = 1) = \frac{E[N^{(3)}]}{1}\big[0.05 \cdot P^{(2)}(S_N = 0)\big] = 8.58532695375(0.05)(0.01137345524) = 0.00488224159.$$

Computing $P^{(s)}(S_N = 2)$:

$$\begin{aligned} P^{(1)}(S_N = 2) &= \frac{E[N^{(1)}]}{2}\big[0.05 \cdot P^{(2)}(S_N = 1) + 0.8 \cdot P^{(2)}(S_N = 0)\big] \\ &= \frac{5.87179487179}{2}\big[0.05(0.00162272948) + 0.8(0.01137345524)\big] = 0.02695124683. \\ P^{(2)}(S_N = 2) &= \frac{E[N^{(2)}]}{2}\big[0.05 \cdot P^{(3)}(S_N = 1) + 0.8 \cdot P^{(3)}(S_N = 0)\big] \\ &= \frac{7.30131004367}{2}\big[0.05(0.00488224159) + 0.8(0.00444503648)\big] = 0.01387300480. \end{aligned}$$

Computing $P^{(1)}(S_N = 3)$:

$$\begin{aligned} P^{(1)}(S_N = 3) &= \frac{E[N^{(1)}]}{3}\big[0.05 \cdot P^{(2)}(S_N = 2) + 0.8 \cdot P^{(2)}(S_N = 1) + 0.3 \cdot P^{(2)}(S_N = 0)\big] \\ &= \frac{5.87179487179}{3}\big[0.05(0.01387300480) + 0.8(0.00162272948) + 0.3(0.01137345524)\big] \\ &= 0.01057680615. \end{aligned}$$

Computing $P^{(0)}(S_N = 4)$:

$$\begin{aligned} P^{(0)}(S_N = 4) &= \frac{E[N^{(0)}]}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\ &= \frac{4.33333333333}{4}\big[0.05(0.01057680615) + 0.8(0.02695124683) \\ &\qquad + 0.3(0.00333912981) + 0.02933480769\big] \\ &= \boxed{0.05679127315}. \end{aligned}$$

Example 9 (N in a Finite Mixture of Poisson, Binomial and Negative Binomial Distributions)**.** Consider a discrete mixture distribution where the event is generated by one of three random processes. The event is either generated by process one, which follows a Poisson distribution with $\lambda = 3$, or process two, which follows a Binomial distribution with $r_2 = 6$ and $p_2 = 0.5$, or process three, which follows a Negative Binomial distribution with $r_3 = 2$ and $p_3 = 0.4$. The probability of process one is $\beta_1^{(0)} = 0.4$. The probability of process two is $\beta_2^{(0)} = 0.3$. The probability of process three is $\beta_3^{(0)} = 0.3$. Notice that $\beta_1^{(0)} + \beta_2^{(0)} + \beta_3^{(0)} = 1$. We want to find $P(S_N = 4)$.

So, the probability mass function for this mixed distribution is

$$P^{(s)}(N = n) = \beta_1^{(s)} \cdot \frac{e^{-\lambda}\lambda^n}{n!} + \beta_2^{(s)} \cdot \binom{r_2 - s}{n} p_2^n (1-p_2)^{(r_2-s)-n} + \beta_3^{(s)} \cdot \binom{n + r_3 + s - 1}{n} p_3^{r_3+s}(1-p_3)^n,$$

for $0 \le n \le \max(r_2 - s,\, r_3 + s)$.

So we have,

$$E[N_1^{(s)}] = \lambda = 3, \quad \text{for all } s.$$

$$E[N_2^{(s)}] = (r_2 - s)\, p_2 = (6 - s)(0.5).$$

$$E[N_3^{(s)}] = \frac{(r_3 + s)(1 - p_3)}{p_3} = \frac{(2 + s)(0.6)}{0.4} = (2 + s)(1.5).$$

Computing the mixing weights at each level:

$$\beta_1^{(0)} = 0.4, \quad \beta_2^{(0)} = 0.3, \quad \beta_3^{(0)} = 0.3.$$

$$E[N^{(0)}] = (0.4)(3) + (0.3)(3) + (0.3)(3) = 3.$$

$$\beta_1^{(1)} = 0.4 \cdot \frac{3}{3} = 0.4.$$

$$\beta_2^{(1)} = 0.3 \cdot \frac{3}{3} = 0.3.$$

$$\beta_3^{(1)} = 0.3 \cdot \frac{3}{3} = 0.3.$$

$$E[N^{(1)}] = 0.4(3) + 0.3(2.5) + 0.3(4.5) = 3.3.$$

$$\beta_1^{(2)} = 0.4 \cdot \frac{3}{3.3} = 0.36363636364.$$

$$\beta_2^{(2)} = 0.3 \cdot \frac{2.5}{3.3} = 0.22727272727.$$

$$\beta_3^{(2)} = 0.3 \cdot \frac{4.5}{3.3} = 0.40909090909.$$

$$
\begin{aligned}
E[N^{(2)}] &= 0.36363636364(3) + 0.22727272727(2) + 0.40909090909(6) = 4. \\
\beta_1^{(3)} &= 0.36363636364 \cdot \frac{3}{4} = 0.27272727273. \\
\beta_2^{(3)} &= 0.22727272727 \cdot \frac{2}{4} = 0.11363636364. \\
\beta_3^{(3)} &= 0.40909090909 \cdot \frac{6}{4} = 0.61363636364. \\
E[N^{(3)}] &= 0.27272727273(3) + 0.11363636364(1.5) \\
&\quad + 0.61363636364(7.5) = 5.59090909091. \\
\beta_1^{(4)} &= 0.27272727273 \cdot \frac{3}{5.59090909091} = 0.14634146341. \\
\beta_2^{(4)} &= 0.11363636364 \cdot \frac{1.5}{5.59090909091} = 0.03048780488. \\
\beta_3^{(4)} &= 0.61363636364 \cdot \frac{7.5}{5.59090909091} = 0.82317073171. \\
E[N^{(4)}] &= 0.14634146341(3) + 0.03048780488(1) \\
&\quad + 0.82317073171(9) = 7.87804878049.
\end{aligned}
$$

Now we can begin the computation to find $P^{(0)}(S_N = 4)$. We want to find,

$$P^{(0)}(S_N = 4) = \frac{E[N^{(0)}]}{4}\big[0.05{\cdot}P^{(1)}(S_N = 3){+}0.8{\cdot}P^{(1)}(S_N = 2){+}0.3{\cdot}P^{(1)}(S_N = 1){+}P^{(1)}(S_N = 0)\big].$$

Base cases:

$$P^{(s)}(N = 0) = \beta_1^{(s)}\, e^{-3} + \beta_2^{(s)}\, (0.5)^{6-s} + \beta_3^{(s)}\, (0.4)^{2+s}.$$

$$P^{(0)}(S_N = 0) = 0.4\, e^{-3} + 0.3(0.5)^6 + 0.3(0.4)^2 = 0.07260232735.$$

$$P^{(1)}(S_N = 0) = 0.4\, e^{-3} + 0.3(0.5)^5 + 0.3(0.4)^3 = 0.04848982735.$$

$$P^{(2)}(S_N = 0) = 0.36363636364\, e^{-3} + 0.22727272727(0.5)^4 + 0.40909090909(0.4)^4 = 0.04278166122.$$

$$P^{(3)}(S_N = 0) = 0.27272727273\, e^{-3} + 0.11363636364(0.5)^3 + 0.61363636364(0.4)^5 = 0.03406647319.$$

$$P^{(4)}(S_N = 0) = 0.14634146341\, e^{-3} + 0.03048780488(0.5)^2 + 0.82317073171(0.4)^6 = 0.01827957098.$$

Computing $P^{(s)}(S_N = 1)$:

$$P^{(1)}(S_N = 1) = \frac{E[N^{(1)}]}{1}\big[0.05 \cdot P^{(2)}(S_N = 0)\big] = 3.3(0.05)(0.04278166122) = 0.00705897410.$$

$$P^{(2)}(S_N = 1) = \frac{E[N^{(2)}]}{1}\big[0.05 \cdot P^{(3)}(S_N = 0)\big] = 4(0.05)(0.03406647319) = 0.00681329464.$$

$$P^{(3)}(S_N = 1) = \frac{E[N^{(3)}]}{1}\big[0.05 \cdot P^{(4)}(S_N = 0)\big] = 5.59090909091(0.05)(0.01827957098) = 0.00510997098.$$

Computing $P^{(s)}(S_N = 2)$:

$$\begin{aligned}
P^{(1)}(S_N = 2) &= \frac{E[N^{(1)}]}{2}\big[0.05 \cdot P^{(2)}(S_N = 1) + 0.8 \cdot P^{(2)}(S_N = 0)\big] \\
&= \frac{3.3}{2}\big[0.05(0.00681329464) + 0.8(0.04278166122)\big] = 0.05703388962. \\
P^{(2)}(S_N = 2) &= \frac{E[N^{(2)}]}{2}\big[0.05 \cdot P^{(3)}(S_N = 1) + 0.8 \cdot P^{(3)}(S_N = 0)\big] \\
&= \frac{4}{2}\big[0.05(0.00510997098) + 0.8(0.03406647319)\big] = 0.05501735420.
\end{aligned}$$

Computing $P^{(1)}(S_N = 3)$:

$$\begin{aligned}
P^{(1)}(S_N = 3) &= \frac{E[N^{(1)}]}{3}\big[0.05 \cdot P^{(2)}(S_N = 2) + 0.8 \cdot P^{(2)}(S_N = 1) + 0.3 \cdot P^{(2)}(S_N = 0)\big] \\
&= \frac{3.3}{3}\big[0.05(0.05501735420) + 0.8(0.00681329464) + 0.3(0.04278166122)\big] \\
&= 0.02313960197.
\end{aligned}$$

Computing $P^{(0)}(S_N = 4)$:

$$\begin{aligned}
P^{(0)}(S_N = 4) &= \frac{E[N^{(0)}]}{4}\big[0.05 \cdot P^{(1)}(S_N = 3) + 0.8 \cdot P^{(1)}(S_N = 2) + 0.3 \cdot P^{(1)}(S_N = 1) + P^{(1)}(S_N = 0)\big] \\
&= \frac{3}{4}\big[0.05(0.02313960197) + 0.8(0.05703388962) + 0.3(0.00705897410) + 0.04848982735\big] \\
&= \boxed{0.07304370853}.
\end{aligned}$$

8. Computational Analysis

Computational analysis was conducted to compare accuracy of results, number of evaluations required, and memory used when utilizing the recursive algorithm defined in Corollary 3 as compared to the convolutional Law of Total Probability method discussed in the first example.

8.1. Accuracy

The computation results of both methods agreed to a level of precision up to the precision of the underlying numeric data structure of the programming language ($\cong 10^{-10}$). This level of accuracy was consistent throughout all the examples. The results were further validated against a Monte Carlo simulation of 500,000 draws simulating the results of each of the examples at various levels of K. All results agreed with the Monte Carlo simulated values within less than 6% relative error. Therefore, the computational analysis provided clear validation of the accuracy of the method.

8.2. Memory Usage

The implementation of both methods requires the use of memoization. Without memoization the recursion tree requires both methods to recompute every subproblem and the number of evaluations grows exponentially. This would make computation untenable at levels of $k \geq 20$. Memory usage was measured as the number of evaluations cached in the process of computing a result. The recursive method required an $O(k^2)$ level of storage to compute a result. The recursive method's evaluations accumulate in memory and are never evicted. The Law of Total Probability method required an $O(k)$ level of storage as only the current convolution slice is held in memory and previous slices are evicted. The Law of Total Probability method had a clear advantage over the recursive method in memory usage.

8.3. Evaluation Efficiency

Evaluation efficiency was measured between methods by counting the number of convolution-sum multiplications. For the recursive method this is the count of $(i \cdot \alpha_i \cdot P_{s+1})$ evaluations computed to arrive at a result. For the Law of Total Probability method this is the count of $(\alpha_i \cdot f_{n-1}(j-i))$ evaluations computed to arrive at a result. Both methods resulted in an evaluation efficiency of $O(k^2)$ level of complexity. The recursive method has a slight advantage at small values of k but the number of evaluations converges to 1 to 1 by $k = 200$.

Overall, the new recursive algorithm defined in Corollary 3 does not offer a computational advantage over the pre-existing Law of Total Probability convolutional method. In fact, it is at a computational disadvantage in that it requires significantly more memory to compute its results efficiently. The value of the recursive method lies in the algebraic result itself. The closed-form weight update and the proof that the framework extends to finite mixtures is not in runtime or memory efficiency.

The implementation and its output can be found at `https://github.com/robrighter/masters-thesis/tree/main/implementation`.

9. Closing Statements

9.1. Overview of Contributions

The primary contribution of this thesis is the extension of the Peköz-Ross identity to the case of finite mixtures of $(a, b, 0)$ class distributions. Corollary 3 derives a more general form of the recursive expression for calculating compound random variables. Furthermore, this corollary provides a closed-form formula for the recursive mixture weights, bypassing the need for nested recursion. This allows the framework to calculate the single-distribution examples in the Peköz and Ross 2004 paper, as well as complex counting distributions made up of finite mixtures of the $(a, b, 0)$ class.

Panjer (1981) and Sundt & Jewell (1981) established the $(a, b, 0)$ class using direct recurrence, and Willmot (1993) extended the recurrence to continuous mixtures. However, those continuous mixtures ultimately collapse into a single marginal distribution (like the Negative Binomial). The finite mixture counting distributions introduced in this thesis maintain the structural integrity of distinct subsets, such as higher-risk versus lower-risk insurance groups.

The technique offers a unified recursive method regardless of which single $(a, b, 0)$ class counting distribution is chosen or finite mixtures of $(a, b, 0)$ class distributions. This reduces some of the complexity associated with manual computation of probabilities. While the method does not introduce a computational advantage over existing convolutional techniques, computation times associated with this recursive method are at parity with existing methods.

9.2. Applications and Future Work

The primary value of the finite mixture framework lies in its ability to model genuine population subgroups in the counting distribution of compound random variables. While prior work often assumes a homogeneous counting distribution for mathematical convenience, real-world data, particularly in insurance and telecommunications, is rarely so uniform.

A clear application is aggregate loss modeling. For example, a car insurance portfolio is naturally composed of distinct subgroups. These might include a 'low risk group' modeled with the Poisson distribution using a small λ, and a 'high risk group' modeled with a Poisson distribution using a large λ. This method provides a recursive tool for calculating the risk probability while maintaining that population segmentation throughout the calculations. The same pattern is applicable in other domains such as healthcare, where one might segment patient populations, or telecommunications networks, where one might segment traffic groups. In each of these cases, modeling the structure of the underlying subgroups with finite mixtures can yield a more accurate probability model.

There are several promising possibilities for future research in this area. Because the finite mixture framework is currently limited to $(a, b, 0)$ class counting distributions, a natural next step would be extending the technique to the $(a, b, 1)$ class. Furthermore, this research did not evaluate numerical resilience at extremely high values of k or with a massive number of mixture components. It is entirely possible that this recursive technique offers a numerical stability advantage over conventional convolution methods at larger scales.

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Biographical Statement

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